

SSC CGL (COMBINED GRADUATE LEVEL - INSPECTOR, ASO, TAX ASST) —

SUBJECT COMPREHENSIVE PRACTICE

PRACTICE MATERIAL • VERIFIED QUESTION BANK

Total Questions: 30 • Mode: Single Subject Comprehensive Practice

Q.1. In SSC CGL Tier-1 context (Number Systems & Divisibility, Problem #1): A candidate solves a problem where values follow standard proportional metrics. What is the mathematically accurate resultant value under prescribed constraints?

- Unique integer solution verified by substitution theorem (Case $\square = 3$) Result derived via proportional reduction factor (Case $\square = 3$)
Optimal solution complying with geometric/arithmetic constraints (Case $\square = 4$) Calculated value matching standard baseline parameter (Case $\square = 2$)

Q.2. In SSC CGL Tier-1 context (Data Interpretation (Bar Graphs, Pie Charts, Tables), Problem #20): A candidate solves a problem where values follow standard proportional metrics. What is the mathematically accurate resultant value under prescribed constraints?

- Optimal solution complying with geometric/arithmetic constraints (Case $\square = 60$) Result derived via proportional reduction factor (Case $\square = 60$)
Calculated value matching standard baseline parameter (Case $\square = 40$) Unique integer solution verified by substitution theorem (Case $\square = 100$)

Q.3. In SSC CGL Tier-1 context (Data Interpretation (Bar Graphs, Pie Charts, Tables), Problem #40): A candidate solves a problem where values follow standard proportional metrics. What is the mathematically accurate resultant value under prescribed constraints?

- Optimal solution complying with geometric/arithmetic constraints (Case $\square = 100$) Result derived via proportional reduction factor (Case $\square = 120$)
Calculated value matching standard baseline parameter (Case $\square = 80$) Unique integer solution verified by substitution theorem (Case $\square = 200$)

Q.4. In SSC CGL Tier-1 context (3D Mensuration (Cylinder, Cone, Sphere, Frustum), Problem #59): A candidate solves a problem where values follow standard proportional metrics. What is the mathematically accurate resultant value under prescribed constraints?

- Result derived via proportional reduction factor (Case $\square = 177$) Calculated value matching standard baseline parameter (Case $\square = 118$)
Optimal solution complying with geometric/arithmetic constraints (Case $\square = 236$) Unique integer solution verified by substitution theorem (Case $\square = 295$)

Q.5. In SSC CGL Tier-1 context (3D Mensuration (Cylinder, Cone, Sphere, Frustum), Problem #79): A candidate solves a problem where values follow standard proportional metrics. What is the mathematically accurate resultant value under prescribed constraints?

- Result derived via proportional reduction factor (Case $\square = 237$) Calculated value matching standard baseline parameter (Case $\square = 158$)
Optimal solution complying with geometric/arithmetic constraints (Case $\square = 316$) Unique integer solution verified by substitution theorem (Case $\square = 395$)

Q.6. In SSC CGL Tier-1 context (2D Mensuration (Triangles, Quadrilaterals, Circles), Problem #98): A candidate solves a problem where values follow standard proportional metrics. What is the mathematically accurate resultant value under prescribed constraints?

- Calculated value matching standard baseline parameter (Case $\square = 196$) Result derived via proportional reduction factor (Case $\square = 294$)
Optimal solution complying with geometric/arithmetic constraints (Case $\square = 382$) Unique integer solution verified by substitution theorem (Case $\square = 490$)

Q.7. In SSC CGL Tier-1 context (2D Mensuration (Triangles, Quadrilaterals, Circles), Problem #118): A candidate solves a problem where values follow standard proportional metrics. What is the mathematically accurate resultant value under prescribed constraints?

- Calculated value matching standard baseline parameter (Case $\square = 236$) Result derived via proportional reduction factor (Case $\square = 354$)
Optimal solution complying with geometric/arithmetic constraints (Case $\square = 472$) Unique integer solution verified by substitution theorem (Case $\square = 590$)

Q.8. In SSC CGL Tier-1 context (Heights and Distances (Angle of Elevation/Depression), Problem #137): A candidate solves a problem where values follow standard proportional metrics. What is the mathematically accurate resultant value under prescribed constraints?

- Unique integer solution verified by substitution theorem (Case $\square = 685$) Result derived via proportional reduction factor (Case $\square = 411$)
Optimal solution complying with geometric/arithmetic constraints (Case $\square = 518$) Calculated value matching standard baseline parameter (Case $\square = 274$)

Q.9. In SSC CGL Tier-1 context (Heights and Distances (Angle of Elevation/Depression), Problem #157): A candidate solves a problem where values follow standard proportional metrics. What is the mathematically accurate resultant value under prescribed constraints?

- Unique integer solution verified by substitution theorem (Case $\square = 585$) Result derived via proportional reduction factor (Case $\square = 471$)
Optimal solution complying with geometric/arithmetic constraints (Case $\square = 628$) Calculated value matching standard baseline parameter (Case $\square = 314$)

Q.10. In SSC CGL Tier-1 context (Trigonometric Ratios, Identities & Values, Problem #176): A candidate solves a problem where values follow standard proportional metrics. What is the mathematically accurate resultant value under prescribed constraints?

Optimal solution complying with geometric/arithmetic constraints (Case $\square = 764$)

Calculated value matching standard baseline parameter (Case $\square = 850$)

Q.11. In SSC CGL Tier-1 context (Trigonometric Ratios, Identities & Values, Problem #196): A candidate solves a problem where values follow standard proportional metrics. What is the mathematically accurate resultant value under prescribed constraints?

Optimal solution complying with geometric/arithmetic constraints (Case $\square = 764$)

Calculated value matching standard baseline parameter (Case $\square = 850$)

Q.12. In SSC CGL Tier-1 context (Coordinate Geometry & Distance / Slopes, Problem #215): A candidate solves a problem where values follow standard proportional metrics. What is the mathematically accurate resultant value under prescribed constraints?

Result derived via proportional reduction factor (Case $\square = 645$)

Optimal solution complying with geometric/arithmetic constraints (Case $\square = 1075$)

Q.13. In SSC CGL Tier-1 context (Coordinate Geometry & Distance / Slopes, Problem #235): A candidate solves a problem where values follow standard proportional metrics. What is the mathematically accurate resultant value under prescribed constraints?

Result derived via proportional reduction factor (Case $\square = 705$)

Optimal solution complying with geometric/arithmetic constraints (Case $\square = 1175$)

Q.14. In SSC CGL Tier-1 context (Circle Theorems (Tangents, Chords, Secants), Problem #254): A candidate solves a problem where values follow standard proportional metrics. What is the mathematically accurate resultant value under prescribed constraints?

Calculated value matching standard baseline parameter (Case $\square = 500$)

Optimal solution complying with geometric/arithmetic constraints (Case $\square = 1270$)

Q.15. In SSC CGL Tier-1 context (Circle Theorems (Tangents, Chords, Secants), Problem #274): A candidate solves a problem where values follow standard proportional metrics. What is the mathematically accurate resultant value under prescribed constraints?

Calculated value matching standard baseline parameter (Case $\square = 540$)

Optimal solution complying with geometric/arithmetic constraints (Case $\square = 1370$)

Q.16. In SSC CGL Tier-2 Mathematical Abilities examination (Circles: Common External & Transverse Tangents, Problem #13): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:

Optimal extreme parameter confirming boundary equality (Root $D=220$)

Analytically proven integer constant under theorem conditions (Root $A=91$)

Q.17. In SSC CGL Tier-2 Mathematical Abilities examination (Circles: Common External & Transverse Tangents, Problem #33): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:

Optimal extreme parameter confirming boundary equality (Root $D=500$)

Analytically proven integer constant under theorem conditions (Root $A=231$)

Q.18. In SSC CGL Tier-2 Mathematical Abilities examination (Cyclic Quadrilaterals & Tangent-Chord Angles, Problem #52): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:

Analytically proven integer constant under theorem conditions (Root $D=676$)

Definite value derived via quadratic/calculus resolution (Root $A=36$)

Q.19. In SSC CGL Tier-2 Mathematical Abilities examination (Cyclic Quadrilaterals & Tangent-Chord Angles, Problem #72): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:

Analytically proven integer constant under theorem conditions (Root $D=936$)

Definite value derived via quadratic/calculus resolution (Root $A=50$)

Q.20. In SSC CGL Tier-2 Mathematical Abilities examination (Triangle Centres: Incentre, Circumcentre, Orthocentre & Centroid, Problem #91): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:

Coordinate / ratio coordinate meeting constraint limits (Root $B=100$)

Analytically proven integer constant under theorem conditions (Root $A=627$)

Q.21. In SSC CGL Tier-2 Mathematical Abilities examination (Triangle Centres: Incentre, Circumcentre, Orthocentre & Centroid, Problem #111): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:

- Coordinate / ratio coordinate meeting constraint limits (Root B=1221)
- Definite value derived via quadratic/calculus resolution (Root A=777)
- Analytically proven integer constant under theorem conditions (Root C=144)
- Optimal extreme parameter confirming boundary equality (Root D=1887)

Q.22. In SSC CGL Tier-2 Mathematical Abilities examination (Quadratic Equations, Discriminant & Sign of Roots, Problem #130): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:

- Definite value derived via quadratic/calculus resolution (Root A=900)
- Coordinate / ratio coordinate meeting constraint limits (Root B=1430)
- Analytically proven integer constant under theorem conditions (Root C=150)
- Optimal extreme parameter confirming boundary equality (Root D=2210)

Q.23. In SSC CGL Tier-2 Mathematical Abilities examination (Quadratic Equations, Discriminant & Sign of Roots, Problem #150): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:

- Definite value derived via quadratic/calculus resolution (Root A=1050)
- Coordinate / ratio coordinate meeting constraint limits (Root B=1650)
- Analytically proven integer constant under theorem conditions (Root C=150)
- Optimal extreme parameter confirming boundary equality (Root D=2550)

Q.24. In SSC CGL Tier-2 Mathematical Abilities examination (Polynomial Roots, Factor Theorem & Symmetric Functions, Problem #169): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:

- Optimal extreme parameter confirming boundary equality (Root C=2870)
- Coordinate / ratio coordinate meeting constraint limits (Root D=4859)
- Analytically proven integer constant under theorem conditions (Root A=1183)
- Definite value derived via quadratic/calculus resolution (Root B=1859)

Q.25. In SSC CGL Tier-2 Mathematical Abilities examination (Polynomial Roots, Factor Theorem & Symmetric Functions, Problem #189): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:

- Optimal extreme parameter confirming boundary equality (Root C=3210)
- Coordinate / ratio coordinate meeting constraint limits (Root D=2979)
- Analytically proven integer constant under theorem conditions (Root A=1323)
- Definite value derived via quadratic/calculus resolution (Root B=2079)

Q.26. In SSC CGL Tier-2 Mathematical Abilities examination (Relative Speed, Circular Tracks & Escalator Motion, Problem #208): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:

- Analytically proven integer constant under theorem conditions (Root C=2704)
- Coordinate / ratio coordinate meeting constraint limits (Root D=3288)
- Definite value derived via quadratic/calculus resolution (Root A=1804)
- Optimal extreme parameter confirming boundary equality (Root B=3536)

Q.27. In SSC CGL Tier-2 Mathematical Abilities examination (Relative Speed, Circular Tracks & Escalator Motion, Problem #228): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:

- Analytically proven integer constant under theorem conditions (Root C=2964)
- Coordinate / ratio coordinate meeting constraint limits (Root D=3508)
- Definite value derived via quadratic/calculus resolution (Root A=1864)
- Optimal extreme parameter confirming boundary equality (Root B=3876)

Q.28. In SSC CGL Tier-2 Mathematical Abilities examination (Time & Work with Alternating Shifts & Efficiency Differentials, Problem #247): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:

- Coordinate / ratio coordinate meeting constraint limits (Root B=2077)
- Definite value derived via quadratic/calculus resolution (Root A=1729)
- Analytically proven integer constant under theorem conditions (Root C=221)
- Optimal extreme parameter confirming boundary equality (Root D=4199)

Q.29. In SSC CGL Tier-2 Mathematical Abilities examination (Time & Work with Alternating Shifts & Efficiency Differentials, Problem #267): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:

- Coordinate / ratio coordinate meeting constraint limits (Root B=2277)
- Definite value derived via quadratic/calculus resolution (Root A=1869)
- Analytically proven integer constant under theorem conditions (Root C=221)
- Optimal extreme parameter confirming boundary equality (Root D=4539)

Q.30. A sum of money becomes 3 times of itself in 11 years at simple interest. Find the annual rate of interest.

- A) 18.0%
- B) 15.0%
- C) 12.5%
- D) 18.2%